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The Facies Impact on Dynamic CO₂ Storage Capacity and Injectivity of Carbonate Reservoir: A Case Study from Saudi Arabia

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Abstract

Saline aquifers provide significant storage capacity for CO₂ geological sequestration, carbonate reservoirs are increasingly considered as potential subsurface candidate for CO₂ storage. However, carbonate reservoirs are more complicated than siliciclastic reservoirs (i.e. sandstone), their complex depositional facies architecture and diagenetic history pose significant challenges for accurate estimation of CO₂ storage capacity, injectivity and containment since different reservoir facies exhibit varied rock types, reservoir properties, heterogeneities and diagenetic histories. This study presents an integrated facies-based workflow applied to carbonate reservoir saline aquifer to assess the impact of reservoir facies on CO₂ sequestration performance. This methodology combines core description, well logs interpretations, facies classification using principal component analysis (PCA), static modelling, and multi-phase flow simulations. The results indicate that peloidal grainstone and peloidal grain-dominated packstone facies have superior storage capacity and injectivity while maintaining favorable pressure profile, while floatstone and coral-dominated facies exhibit high pressure build-up and lower CO₂ injection. This study demonstrates the crucial importance of facies-based modelling for reliable assessment of CO₂ storage and risk management in Carbonate reservoirs.

Introduction

In order to address climate change, significant reduction and removal of Carbon Dioxide (CO₂) from emissions is urgently required. CO₂ capture and storage (CCS) has been proved as one of the critical enabling technologies that could reduce CO₂ emissions significantly while also maintaining the fossil fuels to meet energy requirement and energy transition (IPCC, 2005).

CO₂ sequestration in siliciclastic reservoirs (i.e. sandstones) has been extensively studied and proved by several global CCS projects. In contrast, carbonate reservoirs, due to their geological complexity, remain underutilized and less researched, but this kind of reservoirs could also bring significant potential for CO₂ storage, hence in-depth studies of CO₂ sequestration in carbonate reservoirs are also crucial to mitigate the problems of CO₂ emissions.

It's well-known that the heterogeneity in facies types, pore structures and diagenetic alterations in carbonate reservoirs pose challenges for conventional hydrocarbon fields production. Similarly, it is

necessary to conduct a detailed study to understand the potential and risks of CO₂ storage in carbonate reservoirs. This paper focuses on an Oxfordian-age carbonate reservoir, aiming to demonstrate the significance of facies in assessing CO₂ storage performance and identifying favorable injection zones through a facies-driven integrated static and dynamic modelling.

Geological Settings

The Upper Jurassic (Oxfordian) Hanifa Formation consists of a carbonate succession throughout the outcrop and subsurface of Saudi Arabia (Figure 1). It was deposited in an intra-cratonic passive margin setting, in a shallow marine epeiric platform system, with environments ranging from shallow lagoon, with shoal complex and reef buildups at the margins, to a relatively deep marine carbonates in the organic-rich intrashelf Arabian Basin to the east. The Hanifa formation consists of predominantly carbonate successions and has been interpreted as two 3rd order sequence which can be recognized based on main two flooding surfaces. In the subsurface, its lithology generally consists of a uniform limestone succession, with porous grainstone facies in the upper section and mudstone facies in the lower sequence.

ICC-2020 (Ma)	Era	System/ Period	Series/ Epoch	Stage/ Age	Formation	Member	
						Outcrop	Subsurface
155	Mesozoic	Jurassic	Upper	Kimmeridgian	Jubaila	J2	
						J1	
160				Oxfordian	Hanifa	Ulayyah Member	Upper
						Hawtah Member	Lower

Figure 1—Stratigraphy Column of Late Jurassic in Saudi Arabia, the target formation is Oxfordian Hanifa.

During late Jurassic times, the Arabian Plate was located at tropical latitudes in the southern hemisphere and was part of a passive margin to the south of the Neo-Tethys Ocean (Stampfli, 2000) (Figure 2). The Arabian Plate was tectonically stable during the Jurassic period. The Arabian Platform formed part of a tropical shallow marine epeiric platform system.

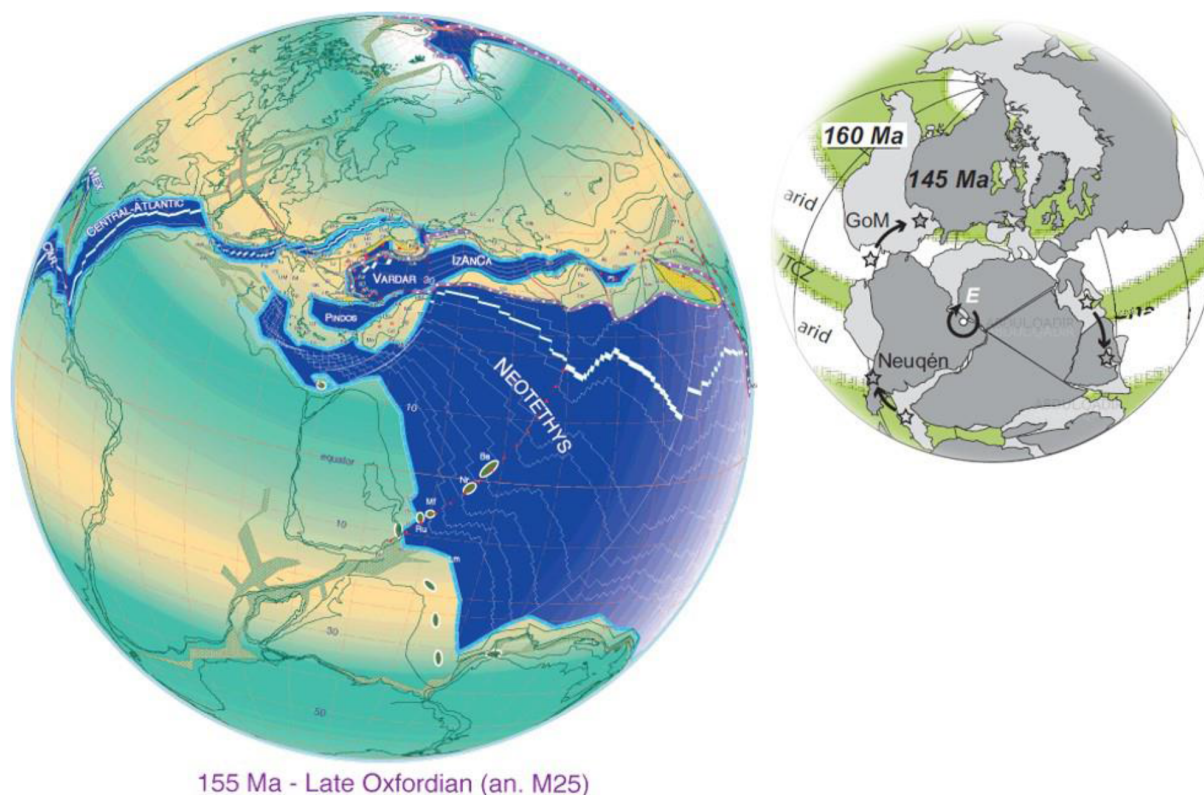


Figure 2—Late Oxfordian to Early Kimmeridgian paleogeography, showing location of the Arabian Plate at tropical paleolatitudes and the opening Neo-Tethys Ocean at the western border.

The Gross Depositional Environments study shows that the Hanifa formation was formed in an overall low-energy inner ramp with high-energy storm-generated events based on interpretation of paleontological data. The Hanifa depositional environment was dominated by argillaceous lime mudstone in mixed carbonate-siliciclastic inner lagoon with relatively deeper settings in the intra-shelf basins, rimmed by the high-energy stromatoporoid shoal complex. Deeper water settings were found in the intra-shelf basins to the east, where low energy environments are encountered below storm wave base.

Controlled by the depositional environment, the Hanifa formation consists of predominantly carbonate succession. The lower part of the Hanifa formation is dominated by calcareous shales/marls, while grainstone facies are more prominent in the upper part of Hanifa. In the type section, the Hanifa formation consists of alternating grey mudstone and brown oolitic grainstone, with little lateral lithological variation.

The Hanifa formation can be divided into two formal members, the lower member's average thickness of 50–60m and consists of argillaceous limestones, interbedded with grainy bioturbated wackestone to packstone beds with rare coral and stromatoporoid boundstone/floatstone. These interbedded cycles of grainy to muddy facies form small cuestas on top of the under-burden cliff. The upper member's thickness ranges from 50–75m and consists of a brown intraclastic grainstone unit at the base and cliff forming reef-dominated boundstone/floatstone beds at the top.

The contact between the upper and lower member was identified by the occurrence of cross-bedded, pelletoid and bioclastic grainstone beds. Vaslet et al. (1983) and Manivit et al. (1985) illustrate that this contact is disconformable in the southern distribution area, accompanied by a layer of carbonate-pebble, conglomeratic base-filling channels into the top of the Upper Member.

Although the lateral reservoir distribution is influenced by tectonic activities, sea-level changes, paleo-structure features and facies variations, the Hanifa formation was penetrated by many oil and gas wells in the region, extending from the western outcrop to the eastern deep subsurface section. The lateral distribution

is extensive but with varying reservoir. This extensive lateral distribution is critical not only for CO₂ sequestration but also for pressure dissipation.

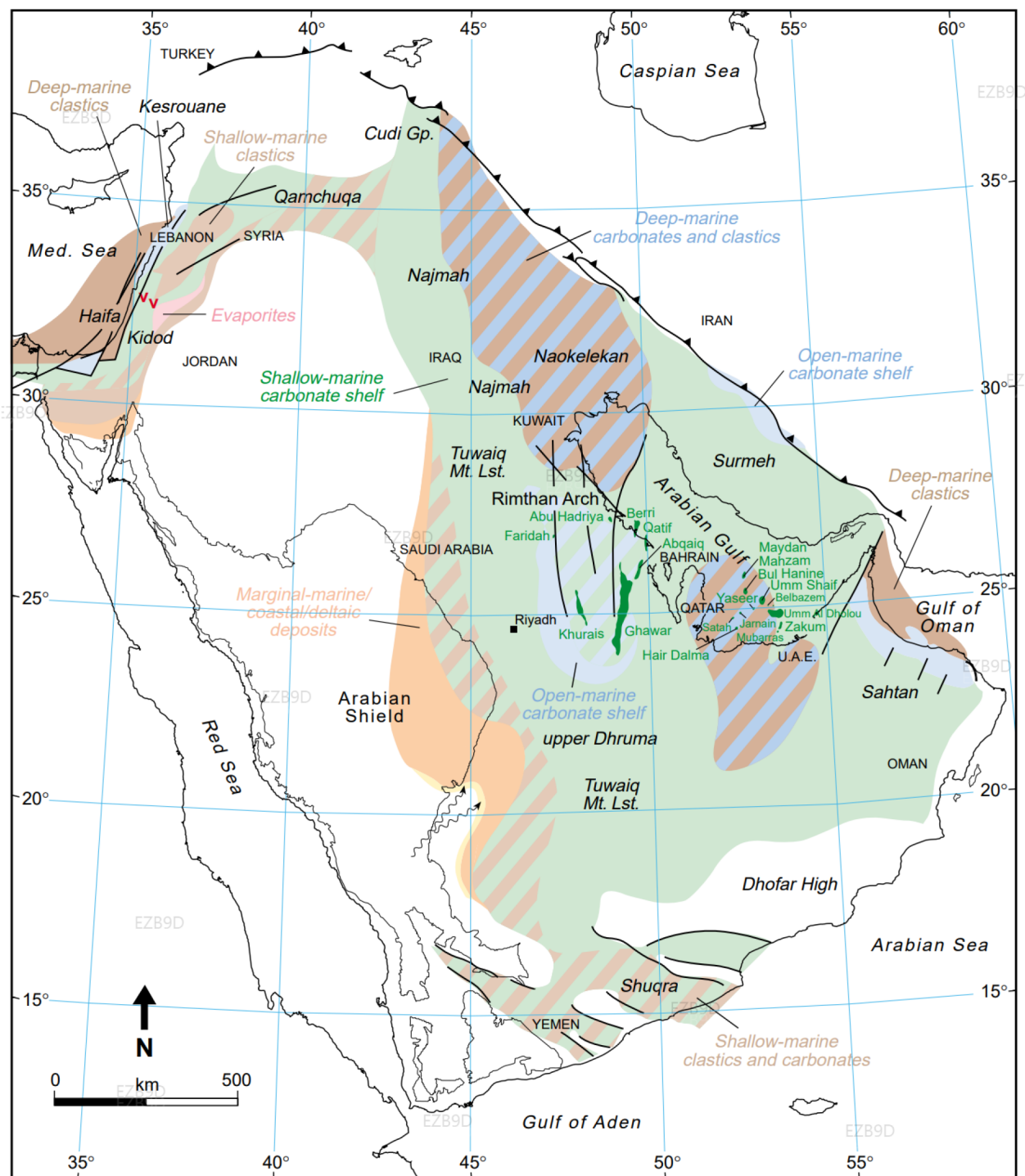


Figure 3—Paleo-facies of the late Middle Jurassic: Callovian to Oxfordian (M.A. Ziegler, 2001).

Methodology

The main process of this newly built workflow includes: 1) detailed core facies descriptions and characterization and fine core facies model generation; 2) log facies classification and clustering using principal component analysis (PCA) method by integrating core facies, well logs and image logs; 3) building

the facies-based porosity-permeability transformation for various facies based on core analysis data; 4) calculation of porosity and permeability logs for all available wells based on using facies-based porosity-permeability transformation; 5) computing the rock mechanics and stress of caprock and reservoir for different facies and calibrating with core data and well testing data; 6) building a fine-grid facies-based 3D static model with various reservoir properties, fluids salinity and stress; 7) after calibration of the dynamic model with well testing data, the facies-based multi-scenarios dynamic cases will be set up to assess the uncertainty ranges of capacity and injectivity and associated risks; 8) the high-graded sweet spots of carbonate reservoir for CO₂ storage will be proposed for further appraisal and de-risking.

This process has been illustrated in an integrated workflow (Figure 4) and implemented in an interest of area (AOI) in Saudi Arabia.

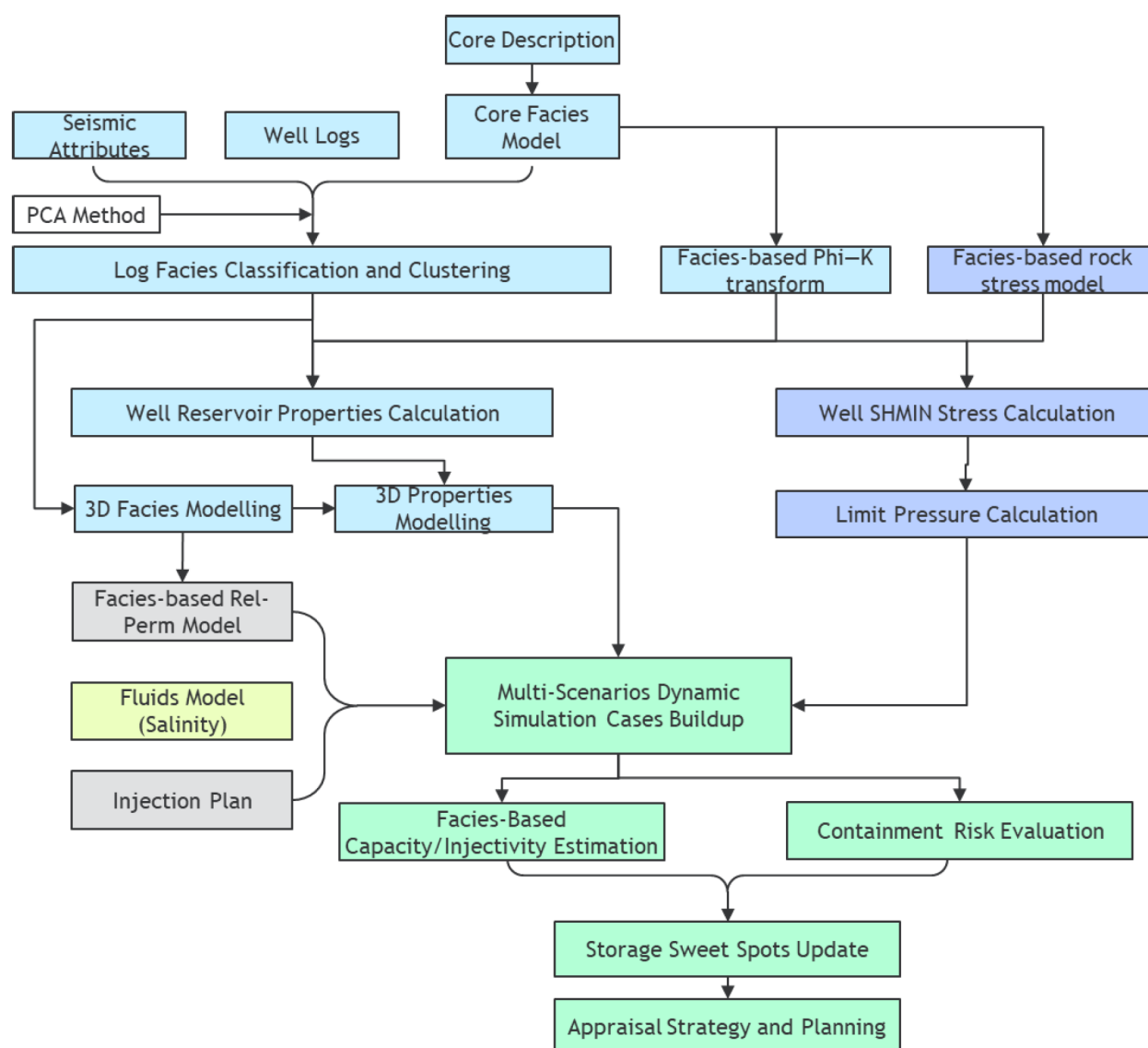


Figure 4—An integrated facies-based 3D static and dynamic modelling workflow for CO₂ storage evaluation.

The AOI is a combination of 4-way structural closure and large surrounding flank area, this closure is confirmed as dry closure by several oil exploration wells. The size of the closure is large with low-relief amplitude, and extended NNW to SSE.

Facies study

Facies and their distribution in space are key building blocks to determine the depositional architectures of storage reservoirs (Zinhom et al., 2010). A comprehensive review of facies and depositional environments has been performed in the area of interest to support the construction of facies-based 3D static and dynamic modelling.

The purpose for this study is to build detail gross depositional environment model to understand and constrain the reservoir and caprock quality, also it will be used as guide to establish the relationship between porosity/permeability and paleo-environment of deposition.

The following data are collected and used to build accurate facies maps for both reservoir and top seal:

- outcrop data
- core and cuttings data
- biostrat analysis data
- well logs
- seismic attributes
- regional geology and analogue

By integration of core analysis and well logs characters, six facies have been recognized and classified on the core and logs based on Dunham Carbonate Classification Scheme, the characters of these identified facies are listed below:

- Peloidal grainstone facies: well-sorted, coarse-grained carbonate rocks composed primarily of peloids, which are small, rounded or oval-shaped grains.
This is the main focus facies for reservoir and predominantly found in high-energy environments, such as shallow marine settings with strong currents or wave action. It is typically located at the platform margins where energy levels are highest, leading to the accumulation of sand-sized peloidal grains.
- Grain-dominated peloidal facies: similar to peloidal grainstone but with a higher concentration of peloids, indicating a slightly different depositional environment.
This is also found in high-energy settings, possibly in areas with even higher energy levels or specific current patterns that favored the accumulation of peloidal grains. It is distributed adjacently to the peloidal grainstone facies, forming a transitional zone.
- Cladocoropsis floatstone facies: contains fossils of Cladocoropsis, a type of bryozoan, mixed with coarse and fine carbonate grains.
It is more localized, suggesting a specific ecological niche. It is found in slightly deeper or more temperate waters where Cladocoropsis could thrive, likely in areas with moderate energy levels and clear water conditions.
- Bioturbated mud-dominated packstone facies: Packstones with sand-sized grains floating in a finer-grained matrix, showing evidence of biological activity such as burrows.
It is indicative of lower energy environments, such as lagoons or deeper marine settings. It is distributed in more protected areas where mud could accumulate and organisms could burrow into the sediment, such as in intra-platform depressions or along the inner platform.
- Stromatoporoid/Coral facies: which includes fossils of stromatoporoids and corals, typical of reef ecosystems.

This is found in shallow, tropical marine environments with clear waters and suitable conditions for reef growth. It is typically located at the platform margins where there is sufficient water depth and energy for reef development.

- Wackestones facies: high proportion of fine-grained matrix with a low proportion of sand-sized grains.

It forms in low-energy environments, such as deep marine basins or protected lagoons, where mud can settle out of the water column without disturbance from strong currents. It is distributed towards the basin, away from the high-energy platform margins.

Furthermore, the facies maps for both reservoir and caprock have been generated, for reservoir interval, the AOI is mainly situated in facies of pelloidic Grain dominated limestone, which exhibit good quality of reservoir, but to the east of the AOI, the facies gradually changed to ooids and pelloid limestone facies mainly due to sea-level fluctuations and structural variations, however, for the overburden top seal section, the AOI is mainly covered by shaly and tight carbonate with streaks of ooids and pelloid limestone.

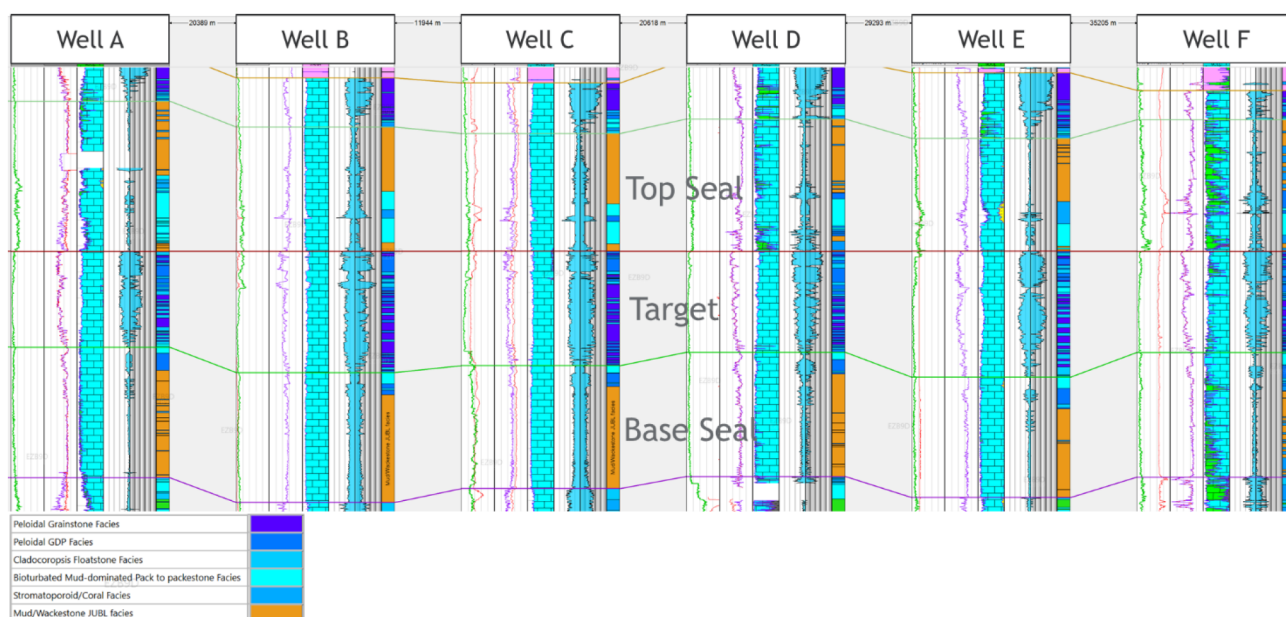


Figure 5—Well correlation across the AOI to illustrate the lateral change of facies. Well log from left to right: GR/CALI, NPHI/RHOB, Lithology, Porosity, Permeability and Facies.

3D static modelling

The static modelling for CCS project is following typical 3D modelling approach, many literatures have explained the process and method of 3D modelling. The key differences highlighted in this paper for 3D static modelling are as below:

- The zone of interval of 3D model covers both reservoir and top/base seal to fully understand the integrity of storage complex.
- Two step facies modelling has been accepted to generate a detail facies model, which will be used as constrain for sequential reservoir properties modelling.

The static model shows that favorable facies with good reservoir quality is mainly located in the lower member of Hanifa formation, and there's a relative extensive buffer zone between lower member and upper member (Figure 6), this could prevent the CO₂ migration or leak upwards. The overburden caprock is mainly in mudstone-wackestone facies, which overall is a competent top seal.

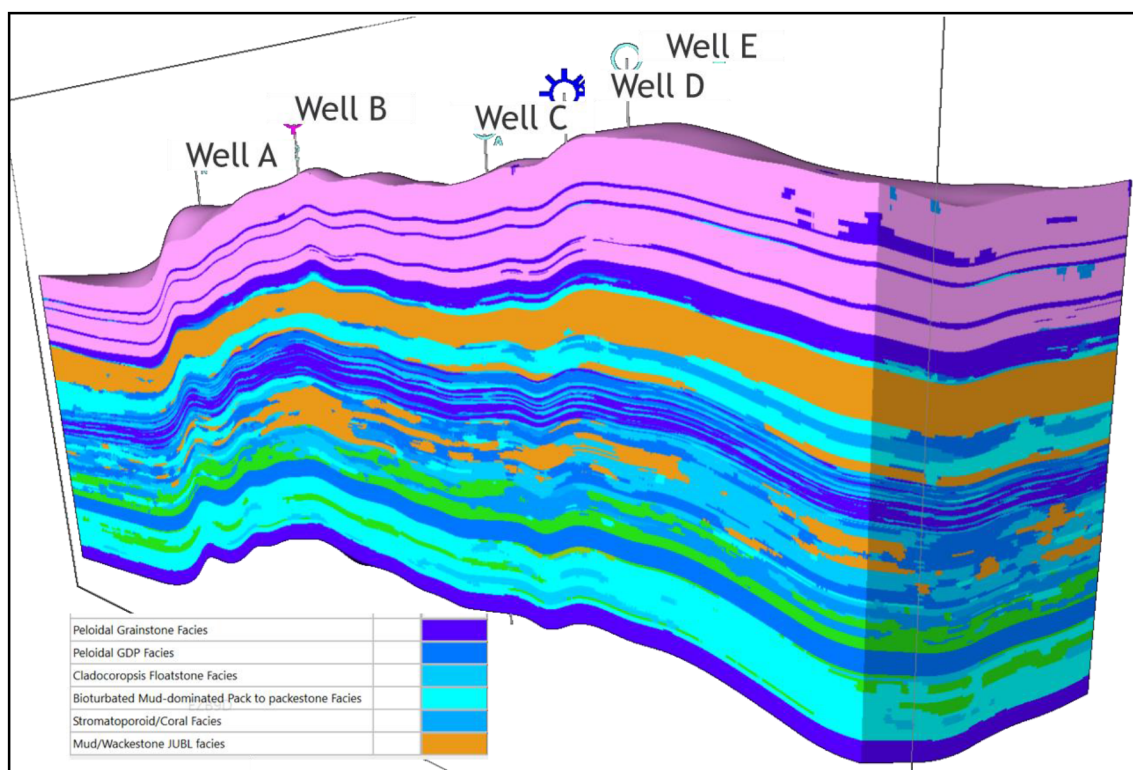


Figure 6—Cross section of 3D facies model

3D dynamic modelling

Facies-based SCAL curves are indispensable for improving the accuracy of carbonate reservoir simulations in CO₂ storage. By accounting for the unique properties of each facies, the SCAL curves enable a more nuanced understanding of CO₂ behavior within the reservoir, hence one key setup for dynamic simulation is to generate the facies-based SCAL curve (Figure 7) based on available core analysis data and analogue data.

From the SCAL curve plot, it's noticed that the grainstone (blue curve) has relatively low critical gas saturation and low residual gas saturation, similar to packstone (pink curve), however for wackestone, it has relatively high residual gas saturation, which means it potentially has more trapped CO₂.

The other key input for dynamic modelling is the caprock stress constraint, which will limit the bottom hole pressure. This study also introduces a 1D geomechanical evaluation based on major litho-facies aiming to understand the potential weakest point of caprock.

A comprehensive sensitivity study has been conducted to evaluate the sensitivity of subsurface parameters. Table 1 lists the parameters. Only the structure and facies have not been considered for this study, but these two parameters could also have high impact on the results.

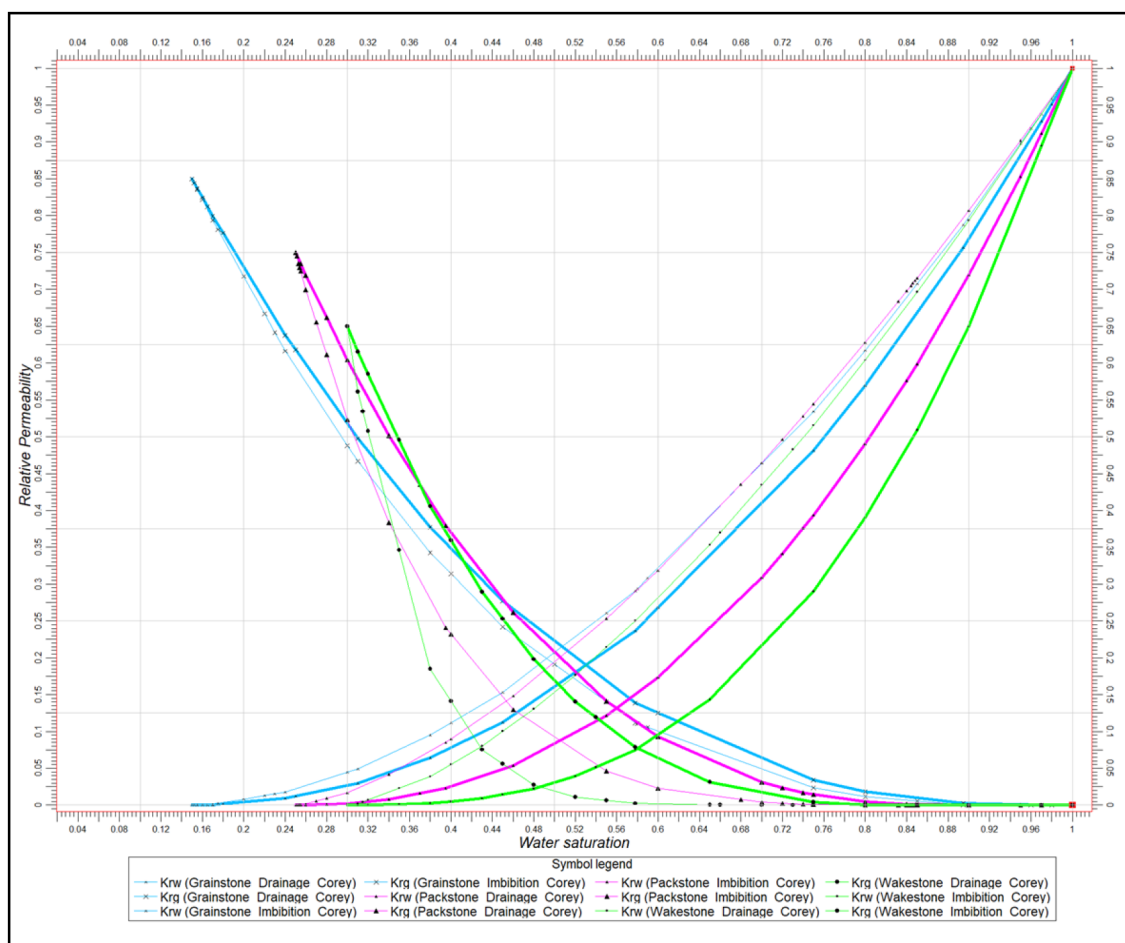


Figure 7—Facies-based SCAL curve for carbonate reservoir

Table 1—Sensitivity Table of Main Subsurface Parameters

#	Parameters	Unit	Low Case	Mid Case	High Case
1	Strucutre	ft	/	Ref Case	/
2	Facies	/	/	Ref Case	/
3	NtG MULT	f	*0.8	1.0	*1.2
4	PHIE MULT	f	-2 PU	1.0	+2 PU
5	PERM MULT	mD	*0.1	1.0	*2.0
6	KvKh	f	0.05	0.1	0.5
7	PV MULT	f	0	50	250
8	Salinity	kPPM	10	20	50
9	ISGCR	f	0.05	0.1	0.35
10	Cr	1/psi	2.00E-06	3.50E-06	5.00E-06
11	SHMIN	psi/ft	0.6	0.7	0.85
12	Pore Pressure	psi/ft	0.4	0.45	0.50
13	Injectivity	MTPA	1	1.5	2

The sensitivity results show that the most sensitive parameters that impact the capacity and injectivity are reservoir permeability, KvKh, residual gas saturation and caprock stress (Figure 8).

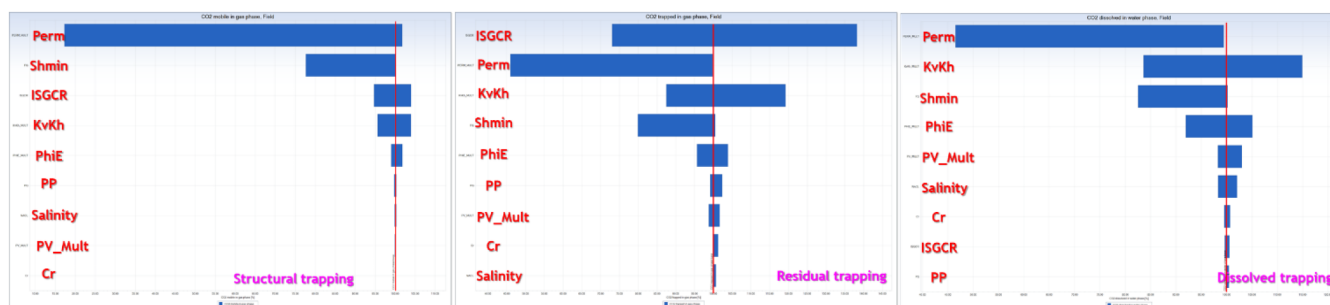


Figure 8—The tornado plot of sensitivity results of key parameters for different trapping. Left: impact on structure trapping; Middle: impact on residual trapping; Right: impact on dissolved trapping

Results

CO₂ Distribution and Storage Capacity

Simulation results indicate that peloidal grainstone and peloidal grain-dominated packstone facies account for over 80% of the total CO₂ injected (Figure 9). These facies exhibit high porosity (10-16%) and permeability (10-200 mD), leading to significant pore volume availability and excellent injectivity.

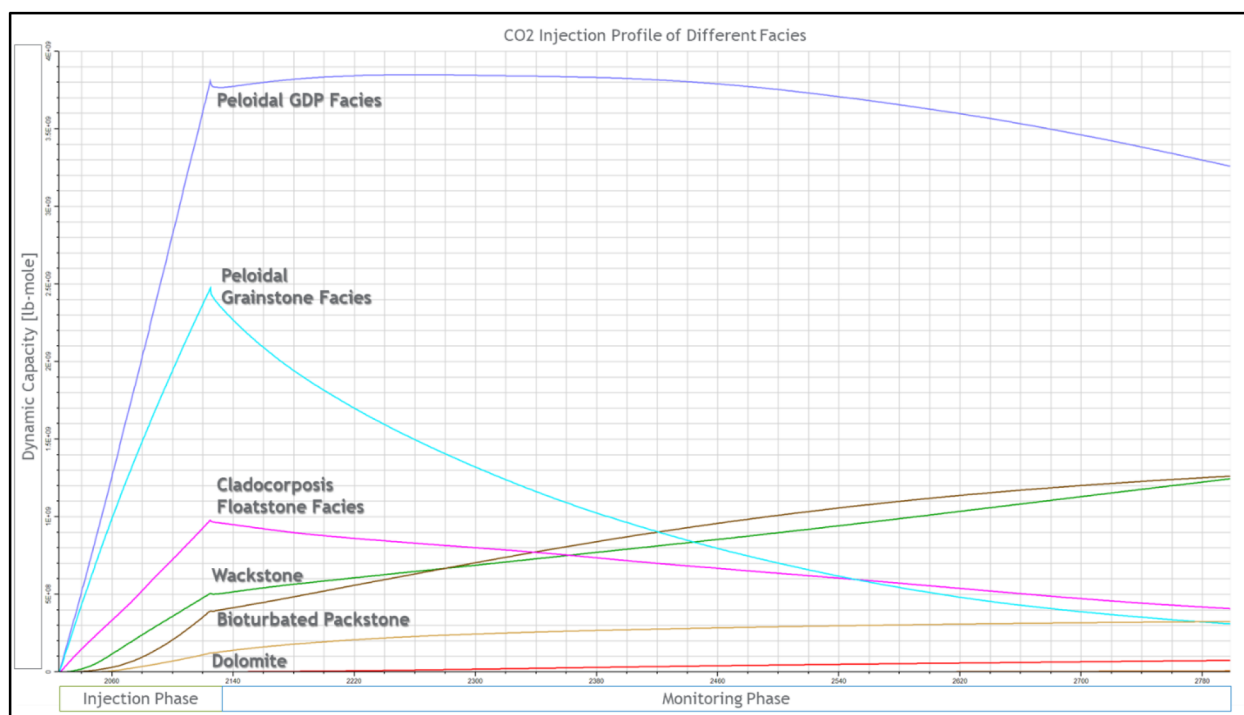


Figure 9—CO₂ injection volume profile for different facies. The peloidal Grain-Dominated Peckstone and Peloidal Grainstone are the top two facies which have high capacity and injectivity.

In contrast, cladocoropsis floatstone and coral/stromatoporoid facies contribute only 12% of CO₂ uptake. These facies are characterized by lower matrix permeability (<5 mD), patchy porosity, and a higher tendency for pressure buildup due to their limited spatial extent and poor connectivity.

Pressure Response and Caprock Integrity

Pore pressure in high-capacity facies increased by only ~11% compared to original formation pressure, indicating safe injectivity margins. However, in floatstone and coral-rich facies, pressure increased by up

to 35%, approaching their estimated fracture pressure thresholds. These areas pose higher risk for caprock breach.

CO₂ Plume Migration

Structural dip and interbedded fractured limestones significantly influence CO₂ plume behavior (Figure 10). The high-permeability fractured zones act as preferential pathways, facilitating rapid plume migration. However, these zones also risk plume leakage if intersecting faults or thin caprock intervals.

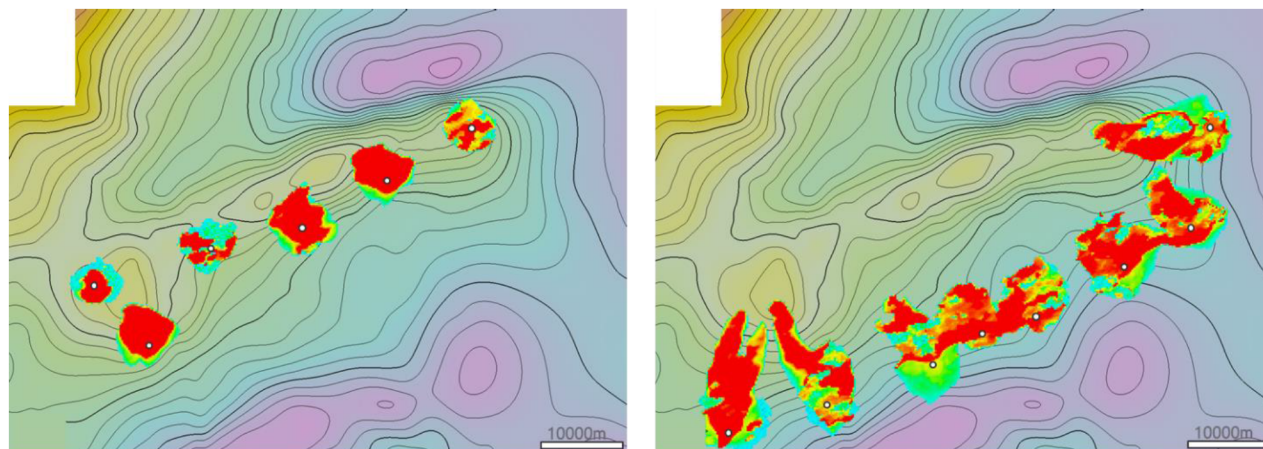


Figure 10—CO₂ plume migration for different location with different facies. Left: injectors close to closure, the plume is constrained by the localized structure; Right: injectors at the flank area, the plume migrates far due to buoyance effect.

Injectivity and Productivity Index

Injectivity indices vary significantly by facies. Peloidal grainstone exhibits the highest injectivity ($>80 \text{ m}^3/\text{day/bar}$), while coral-rich facies show much lower injectivity ($<8.5 \text{ m}^3/\text{day/bar}$). This variability reinforces the need for facies-based well placement strategies.

Discussion

Facies Impact

Grainstone and packstone facies, characterized by interparticle porosity and high permeability, are optimal for CO₂ injection. In contrast, coral-rich facies, with moldic and microporosity, exhibit poor injectivity.

Geomechanical Risks

High-pressure buildup in tight facies suggests that injection wells should avoid these intervals unless assisted by artificial stimulation.

Implications for CCS Planning:

Facies-based screening is crucial for guiding well placement, surveillance planning, and containment risk assessment in carbonate reservoirs.

Conclusion

This study demonstrates a robust, facies-based workflow for dynamic CO₂ storage assessment in a carbonate reservoir. Key conclusions include:

- Facies strongly control CO₂ capacity and injectivity. Peloidal grainstone and packstone are the most favorable facies, offering both high injectivity and safe containment.

- Facies affect pressure evolution. Low-permeability facies may reach near-fracture pressures under moderate injection rates, posing integrity risks.
- Structural and lithological heterogeneity governs plume migration. Fractured limestones can accelerate plume spread but also introduce leakage risks.
- Integrated workflows enhance storage reliability. Combining geology, petrophysics, geomechanics, and reservoir simulation is essential for successful CO₂ sequestration in carbonates.

These findings provide actionable insights for selecting, appraising, and managing CO₂ storage projects in carbonate reservoirs globally, especially in regions like the Middle East where carbonate geology predominates.

Future Work

Even though the CO₂ storage in Carbonate reservoirs is getting increasingly interesting to study, it's still in the early stage and lacking real injection data to calibrate the model, the future research should focus on:

- Coupled geomechanical-fluid simulations to further explore fracture risks.
- Long-term geochemical interactions between CO₂, brine, and carbonate matrix.
- Field pilot testing to validate model predictions and optimize injection strategies.
- Application to other carbonate settings for broader generalization.

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